

chapter 10

Reporting and
Interpreting Bond
Securities

Financial Accounting

12e

Libby • Libby • Hodge

Learning Objectives

After studying this chapter, you should be able to:

- 10-1** Describe the characteristics of bond securities.
- 10-2** Report bonds payable and interest expense for bond securities issued at par.
- 10-3** Compute and analyze the times interest earned ratio.
- 10-4** Report bonds payable and interest expense for bond securities issued at a discount.
- 10-5** Report bonds payable and interest expense for bond securities issued at a premium.
- 10-6** Report the early retirement of bond securities.
- 10-7** Compute and analyze the debt-to-equity ratio.
- 10-8** Explain how bond securities are reported on the statement of cash flows.

Understanding the Business

A company's capital structure is the mixture of **debt** and **equity** it uses to finance its operations.

- Companies raise money from investors by issuing debt securities (referred to as “bonds”) in the bond markets.
 - Borrowing a large amount from an individual bank is often impractical, so companies and governments issue bond to the investing public instead.
- After bonds are issued, they are traded on exchanges such as the New York Bond Exchange.
 - The ability to sell a bond is an advantage to investors because it provides them with liquidity (the ability to convert investments to cash).
 - By issuing more liquid debt that investors can easily buy and sell in the bond markets, companies are able to reduce the cost of long-term borrowing.

Exhibit 10.1 Excerpts from Amazon's Long-Term Debt Footnote

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

Note 6-Debt

	Maturities	December 31, 2022	December 31, 2023
2014 Notes issuance of \$6.0 billion	2024-2044	4,000	4,000
2017 Notes issuance of \$17.0 billion	2024-2057	16,000	15,000
2020 Notes issuance of \$10.0 billion	2025-2060	10,000	9,000
2021 Notes issuance of \$18.5 billion	2024-2061	18,500	17,500
April 2022 Notes issuance of \$12.8 billion	2024-2062	12,750	12,750
December 2022 Notes issuance of \$8.3 billion	2024-2032	8,250	8,250
Credit Facility		1,042	682
Total face value of long-term debt		70,542	67,182
Unamortized discount and issuance costs, net		(393)	(374)
Less: Current portion of long-term debt		(2,999)	(8,494)
Long-term debt		<u>\$ 67,150</u>	<u>\$ 58,314</u>

Learning Objective 10-1

10-1 Describe the characteristics of bond securities.

Characteristics of Bonds Payable - Reasons Why Companies Issue Bonds (1 of 2)

Advantages of bonds

- **Stockholders maintain control** because bondholders do not vote or share in any dividends or payments.
- **A portion of interest expense is tax deductible** which reduces the net cost of borrowing.
- **The return to shareholders can be positive** if the company borrowed at a lower interest rate and invested in projects that earn a higher rate.

Disadvantages of bonds

- **Risk of bankruptcy** exists because the interest payments must be paid each year whether the company earns income or suffers a loss.
- **Negative impact on cash flows** because bonds must be repaid at a specific time in the future. Company must be able to repay the debt or finance it.

Characteristics of Bonds Payable - Reasons Why Companies Issue Bonds (2 of 2)

Assume a company has shareholders' equity of \$100,000 and earns net income of \$20,000 per year. Management plans to spend \$100,000 for new assets expected to earn a net annual \$20,000 per year. To fund the purchase, management **issues bonds or borrow the money** at an interest rate of 8 percent? Assuming the company's debt meets its tax deductibility, the following analysis shows borrowing the money will increase the return to the company's shareholders.

	Option 1 Issue Stock	Option 2 Borrow
Income Statement Impact:		
Income before interest and taxes		\$ 40,000
Interest expense (8% × \$100,000)		8,000
Income before taxes	40,000	32,000
Income taxes expense	8,400	8,000
Net income	<u>\$ 31,600</u>	<u>\$ 24,000</u>
Balance Sheet Impact:		
Stockholders' equity (existing)	\$100,000	\$100,000
Stockholders' equity (new)	\$200,000	\$100,000
Bonds payable		\$100,000
Return on equity (net income/stockholders' equity)	15.80%	25.28%

Bond Terminology (1 of 2)

A bond usually requires the payment of **interest** over its life with repayment of **bond principal** on the maturity date.

The **bond principal** is:

- (1) the amount a company must pay to bondholders at the maturity date
 - (2) the amount used to compute the bond's periodic cash interest payments.
- Bond principal is also called **face value**, **par value**, or **maturity value**.
 - All bonds have a face value. The face value is usually \$1,000, but it can be any amount.



Bond Terminology (2 of 2)

The **coupon rate** is the interest rate specified on a bond, and the rate used to compute the bond's periodic cash interest payment.

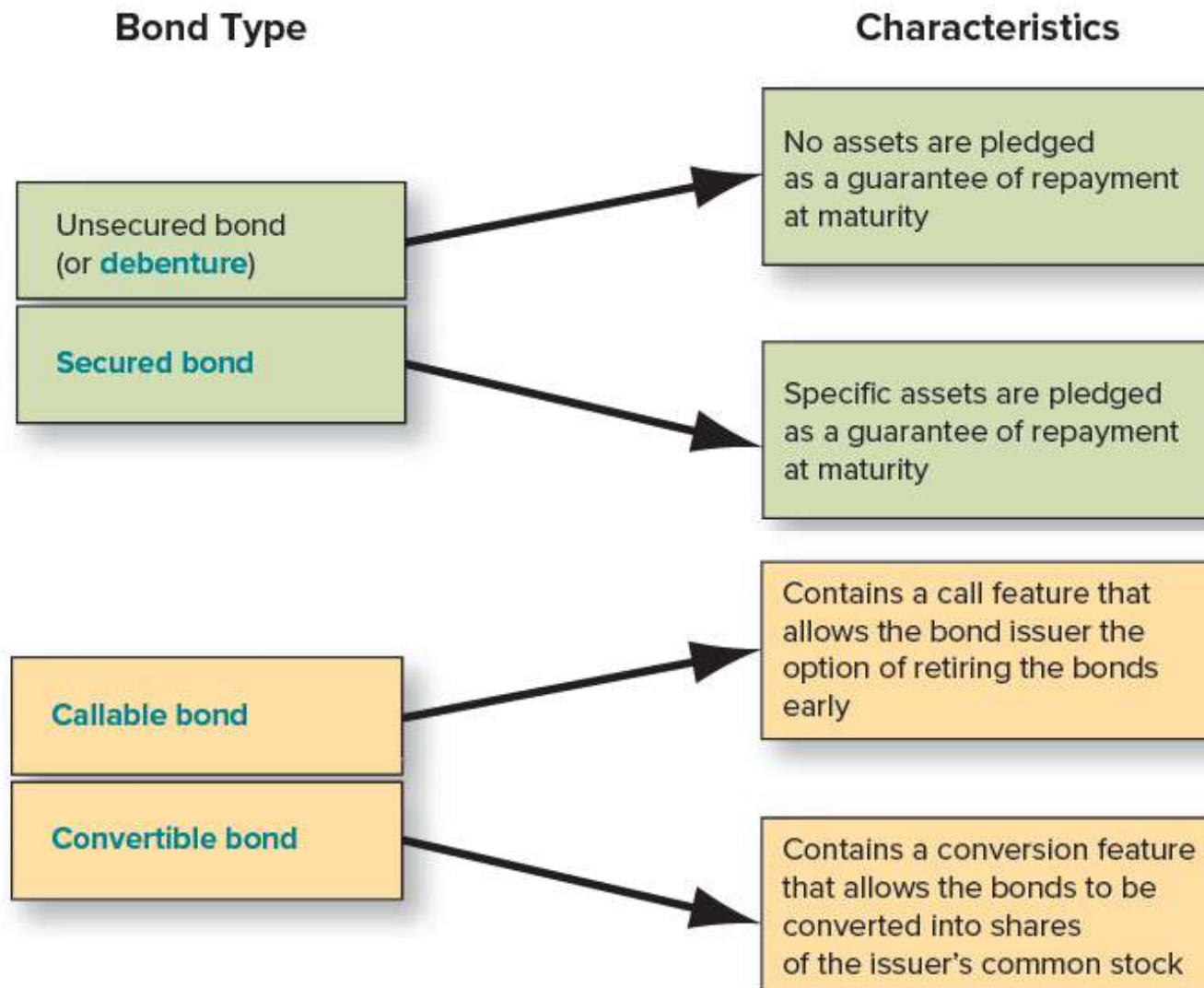
A bond always specifies the coupon rate and the frequency of periodic cash interest payments.

- The **coupon rate** is also called the **stated rate, contract rate, or nominal rate**.
- The interest payments are sometimes called coupon payments.
- A bond's coupon rate is always stated in annual terms.
 - If interest is paid annually the cash interest payment is computed as the bond's face value $\times$ the coupon rate.
 - If the interest payment is made more frequently, computing the cash interest payment requires that the coupon rate be converted to a rate per interest period before it is multiplied by the bond's face value.

Frequency of Interest Payment	Interest Rate per Interest Period	Cash Payment per Interest Period
Annual (once per year)	$8\% \times 1 = 8\%$	$\$1,000 \times 8\% = \80
Semiannual (twice per year)	$8\% \times 1/2 = 4\%$	$\$1,000 \times 4\% = \40
Quarterly (four times per year)	$8\% \times 1/4 = 2\%$	$\$1,000 \times 2\% = \20

Characteristics of different types of bonds

Different types of bonds have different characteristics because individual investors have different risk and return preferences.



Bond Issuance Process

- When a company decides to issue bond securities, it prepares a bond indenture and a bond prospectus.
 - The **indenture** is a legal document that specifies all the details of the bond offering.
 - The **prospectus** is a regulatory document that is filed with the Securities and Exchange Commission. It specifies the details of the bond offering such as: maturity date, rate of interest to be paid, date of each interest payment, other characteristics such as if the bond is callable or convertible.
- The prospectus also describes any **covenants** designed to protect the bondholders such as:
 - limitations on new debt that the company might issue in the future
 - limitations on the payment of dividends to shareholders
 - requirements that the company maintain certain minimum accounting ratios, such as the debt-to-equity ratio.
- The prospectus and annual report also describe how the proceeds of the bond will be used.
- When a bond is issued to an investor, the investor receives a **bond certificate**.
- An independent party, called the **trustee**, is usually appointed to represent the bondholders.

Bond Rating Agencies and Their Assessments of Default Risk



FINANCIAL ANALYSIS

Agencies, such as Standard & Poor's, Moody's, and Fitch evaluate the bond's **default risk** (the risk the bond issuer will not be able to meet the requirements in the prospectus).

- Higher quality bonds have a lower default risk.

Standard & Poor's	Moody's	Fitch	Description	Risk
AAA	Aaa	AAA	Highest investment grade	Low risk ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ High risk
AA	Aa	AA		
A	A	A		
BBB	Baa	BBB	Lowest investment grade	
BB	Ba	BB	Highest junk bond grade	
B	B	B		
CCC	Caa	CCC		
CC	Ca	CC		
C	C	C		
D	C	DDD	In default or unrated	

Reporting Bond Transactions - cash payments (1 of 2)

There are two types of cash payments in the bond contract

1. **Principal.** a single payment made at the end of the bond's life. It is also called
 - **face value**
 - par value
 - maturity value.
2. **Cash interest payments** (also called coupon payments). These payments represent an annuity and are computed as follows:

Face Value	×	Annual Coupon Rate	×	Number of Months/ 12 Months	=	Interest for the Period
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The bond contract specifies whether interest is paid quarterly, semiannually, or annually. Be sure and **use the interest rate per period** in your calculations.

Reporting Bond Transactions - cash payments (2 of 2)

Consider a bond with the following characteristics:

- Face value: \$1,000
- Term: 10 years
- Coupon rate: 6% (annual)
- Interest paid: Semiannually (twice a year)

If you are asked to compute the cash interest payment for this bond, you will need to use an interest rate per period of **3%** (the **semiannual** coupon rate).

Cash Interest Payment Computation						
Face Value	×	Annual Coupon Rate	×	Number of Months/ 12 Months	=	Interest for the Period
\$1,000	×	0.06	×	6/12 months	=	\$30

Coupon interest rate per period = 3%

Market Rate of Interest

- Investors demand a certain rate of return to compensate them for the risks related to the bond offering. The **demanded rate of return** is called the **market interest rate** (also known as the **yield** or **effective interest rate**).
- The issuing company does not determine the price of the bond. The price at which the bonds will sell is determined by the market using a present value calculation. *The market interest rate should be used in computing the present value of a bond.*
- **As market interest rates rise, the price of a bond falls** and vice versa. This is because of the important role the market interest rate plays in determining the present value of a bond. The relationship between the market interest rate and the price of a bond is shown below.

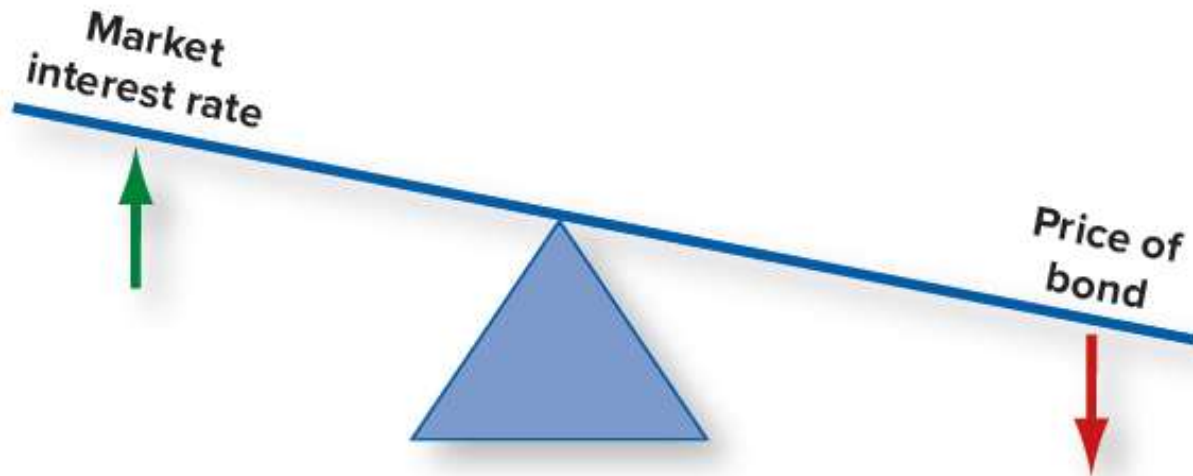


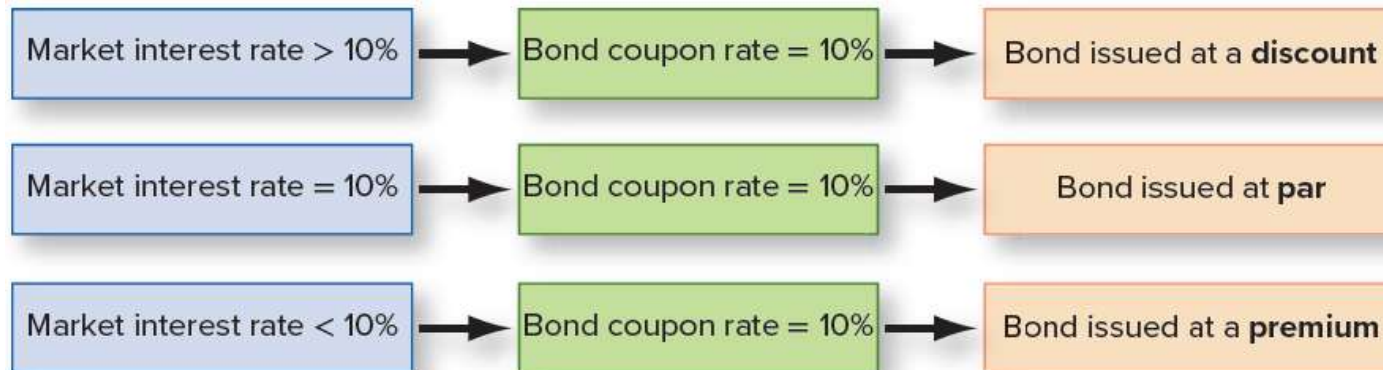
EXHIBIT 10.2

Relationship of Market Interest Rates and Prices of Bonds

Relationship between Coupon Rate and Market Rate (1 of 2)

On the day a company issues a bond, the market interest rate will be either the same as the coupon rate, greater than the coupon rate, or less than the coupon rate.

The relationship between the market interest rate and the bond's coupon rate determines whether the bond is issued at **par**, at a **premium**, or at a **discount**.



Relationship between Coupon Rate and Market Rate (2 of 2)

- When a bond's coupon rate is **less** than the rate investors demand, investors will not buy the bond unless its price is reduced (i.e., a **discount** must be provided).
- When a bond's coupon rate is **more** than investors demand, investors will be willing to pay a **premium** to buy the bond.

Important: Whether a bond is issued at par, at a discount, or at a premium, **investors will always earn the market rate of return.**



A bond's coupon rate relative to the market interest rate determines if a bond is issued at a premium, a discount, or at par.

- During the life of the bond, its market price will change as market interest rates change. This change does not affect the company's financial statements and the way its interest payments are accounted for.
- The interest rates that matter for accounting purposes are the bond's **coupon rate** (which does not change over time) and the **market interest rate on the day the bond is issued**.

Using Artificial Intelligence (AI) to Invest in Bonds



DATA ANALYTICS

- Investors spend considerable time and effort gathering and assessing vast amounts of market information, including market interest rates, general economic conditions, and company-specific information.
- **Using technology to expand the scope of this process would allow investors to make more informed, quicker decisions.**
- AllianceBernstein, a leading investment management and research firm, has created a virtual assistant powered by artificial intelligence that helps investors make bond market decisions.
- You can learn more about Abbie's abilities here:
alliancebernstein.com/fixed-income/bonds/abbie.html

Learning Objective 10-2

10-2 Report bonds payable and interest expense for bond securities issued at par.

Bonds Issued at Par (1 of 5)

Bonds sell at their face (par) value when the **market interest** rate that investors demand is **equal to** the interest rate stated in the bond contract (the **coupon rate**).

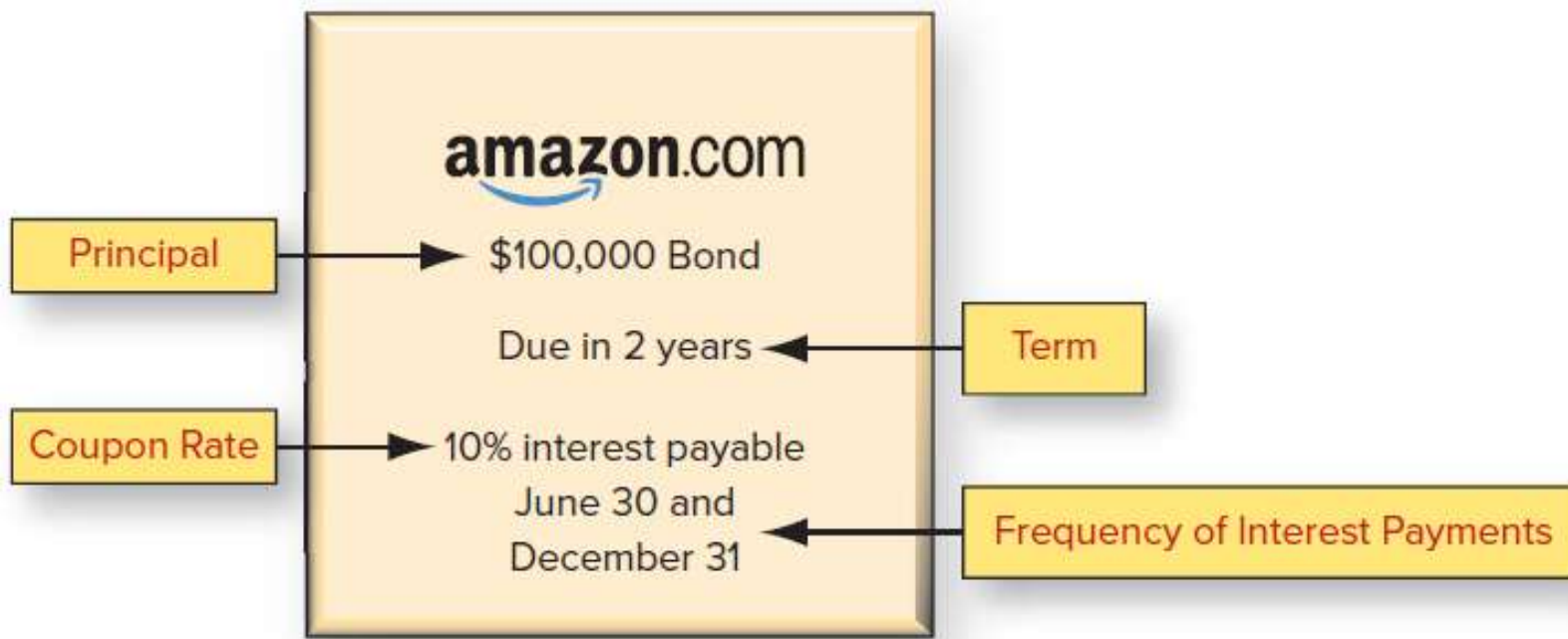
Bonds Issued at Par



Bonds Issued at Par (2 of 5)

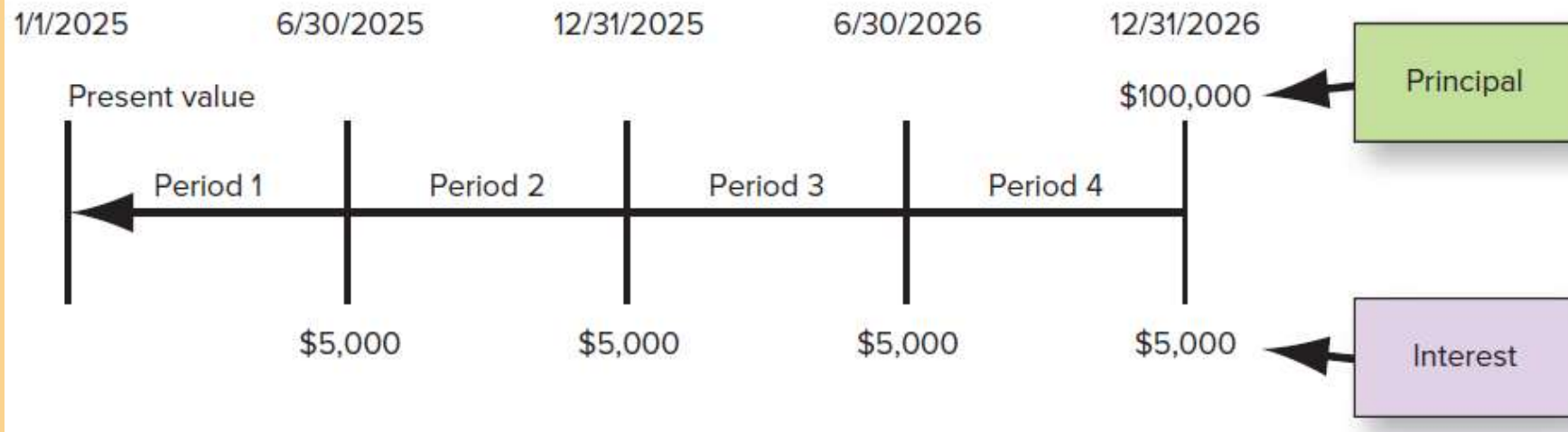
On January 1, 2025, Amazon issues bonds with a coupon rate of 10 percent and a face value of \$100,000. The bonds start accruing interest on January 1, 2025, and will pay interest each June 30 and December 31. The bonds mature in two years on December 31, 2026.

Investors are willing to pay \$100,000 in cash for the bonds meaning the bonds are sold at **Par**.



Bonds Issued at Par (3 of 5)

The amount of money a company receives when it sells bonds is the **present value of the future cash flows** associated with the bonds.



Amazon agrees to make two types of payments in the future: a single payment of \$100,000 when the bond matures in two years and an annuity of \$5,000 [$\$100,000 \times (10\% \times \frac{1}{2} \text{ year})$] payable twice a year for two years (four total payments).

➤ The 10% in the equation is the **bond's coupon rate**.

Bonds Issued at Par (4 of 5)

Use the bond's **market** interest rate per period (in this case $10\% \div 2 = 5\%$) to compute the bond's present value.

The present value of the bond payments can be computed using tables in Appendix E, a financial calculator, or Excel. This calculation uses the present value tables:



Remember: A bond's selling price is determined by its present value, which is computed using the market interest rate per period, not the bond's coupon rate.

Present Value	
Single principal payment at maturity: $\$100,000 \times 0.82270$	\$ 82,270
+ Annuity cash interest payment: $\$5,000 \times 3.54595$	17,730
= Issue price (proceeds from sale) of the bonds	<u>\$100,000</u>

INPUTS USING:

Table E.1	Table E.2	Calculator	Excel
Interest rate (i) = 5%	Interest rate (i) = 5%	Rate (i) = 5	Rate (i) = 0.05
Periods (n) = 4	Periods (n) = 4	Periods (n) = 4	Nper (n) = 4
Factor = 0.82270	Factor = 3.54595	Pmt = $-\$5,000$	Pmt = $-\$5,000$
		FV = $-\$100,000$	Fv = $-\$100,000$

Bonds Issued at Par (5 of 5)

- When the market rate of interest equals the coupon rate, the present value of the future cash flows associated with a bond always equals the bond's face value amount.
- A bond's selling price is determined by the present value of its future cash flows, not the face value.

On the date the bonds are issued, record a bond liability equal to the amount investors are willing to pay for the bonds:

January 1, 2025	Debit	Credit
Cash (+A)	100,000	
Bonds payable (+L)		100,000

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	+100,000		Bonds payable	+100,000			

The amount of interest each period will be \$5,000 ($\$100,000 \times 0.10 \times \frac{1}{2}$ year).

June 30 and December 31 of each year	Debit	Credit
Interest expense (+E, -SE)	5,000	
Cash (-A)		5,000

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	-5,000					Interest expense (+E)	-5,000

Reporting Interest Expense

- Interest expense is reported on the income statement. Because interest is related to financing activities rather than operating activities, it is normally not included in operating expenses on the income statement.
- Interest expense is typically reported just below “income from operations” on the income statement.

AMAZON.COM, INC.			
CONSOLIDATED STATEMENTS OF OPERATIONS (Partial)			
(in millions)			
	Year Ended December 31		
	2021	2022	2023
...			
Operating income	24,879	12,248	36,852
Interest income	448	989	2,949
Interest expense	(1,809)	(2,367)	(3,182)
Other income (expense), net	14,633	(16,806)	938
Total non-operating income (expense)	13,272	(18,184)	705
Income (loss) before income taxes	38,151	(5,936)	37,557

Learning Objective 10-3

10-3 Compute and analyze the times interest earned ratio.

Times Interest Earned

KEY RATIO ANALYSIS



- This ratio shows whether a company is generating sufficient resources from its profit-making operations to meet its current interest obligations.
- A high ratio indicates a large margin of protection. If case profitability deteriorates.
- Failure to meet required interest payments could result in bankruptcy.

$$\text{Times Interest Earned} = \frac{\text{Net Income} + \text{Interest Expense} + \text{Income Tax Expense}}{\text{Interest Expense}}$$

The 2023 ratio for Amazon (dollars in millions):

$$[\$30,182 + \$7,120] / \$2,365 = 12.80$$

COMPARISON OVER TIME		
Amazon		
2022	2022	2023
	(1.51)	12.80

COMPARISON WITH COMPETITORS	
eBay	Walmart
2023	2023
15.06	11.86

Learning Objective 10-4

10-4 Report bonds payable and interest expense for bond securities issued at a discount.

Bonds Issued at a Discount (1 of 4)

Bonds sell at a **discount** when the **market interest rate is greater than the coupon rate**.

Bonds Issued at a Discount



Bonds Issued at a Discount (2 of 4)

On January 1, 2025, Amazon issues bonds with a coupon rate of 10 percent and a face value of \$100,000. The **market interest rate is now 12 percent**. The bonds start accruing interest on January 1, 2025, and will pay interest each June 30 and December 31. The bonds mature in two years on December 31, 2026. The coupon rate (10 percent) is less than the market interest rate (12 percent) on the date of issuance, so the bonds sell at a **discount**.

	Present Value
Single principal payment at maturity: $\$100,000 \times 0.79209$	\$79,209
+ Annuity cash interest payment: $\$5,000 \times 3.46511$	17,326
Issue (sale) price of bonds	<u>\$96,535</u>

Thus, if the market interest rate is 12 percent on the date of issuance, investors will be willing to pay Amazon \$96,535 for the bonds. Some people refer to this price as 96.5, which indicates the selling price was 96.5 percent of the bond's face value ($96.5 = \$96,535 \div \$100,000$).

INPUTS USING:

Table E.1	Table E.2	Calculator	Excel
Interest rate (i) = 6%	Interest rate (i) = 6%	Rate (i) = 6	Rate (i) = 0.06
Periods (n) = 4	Periods (n) = 4	Periods (n) = 4	Nper (n) = 4
Factor = 0.79209	Factor = 3.46511	Pmt = $-\$5,000$	Pmt = $-\$5,000$
		FV = $-\$100,000$	Fv = $-\$100,000$

Bonds Issued at a Discount (3 of 4)

Since the present value of the bonds of \$96,535 is less than the \$100,000 face amount of the bonds, the bonds are issued at a discount of \$3,465.

Companies can account for the bond discount explicitly (using a bond discount account) or implicitly. Regardless of the method used, the dollar value reported on the balance sheet (the bond payable book value) is identical.

January 1, 2025	Debit	Credit
Cash (+A)	96,535	
Bond discount (–L)	3,465	
Bonds payable (+L)		100,000

Bond discount is a contra-liability account

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	+96,535		Bonds payable	+100,000			
			Bond discount	–3,465			

Bonds payable book value: **\$96,535** (\$100,000 beginning balance – \$3,465)



The bond payable book value is equal to the bond's face value less the unamortized bond discount.

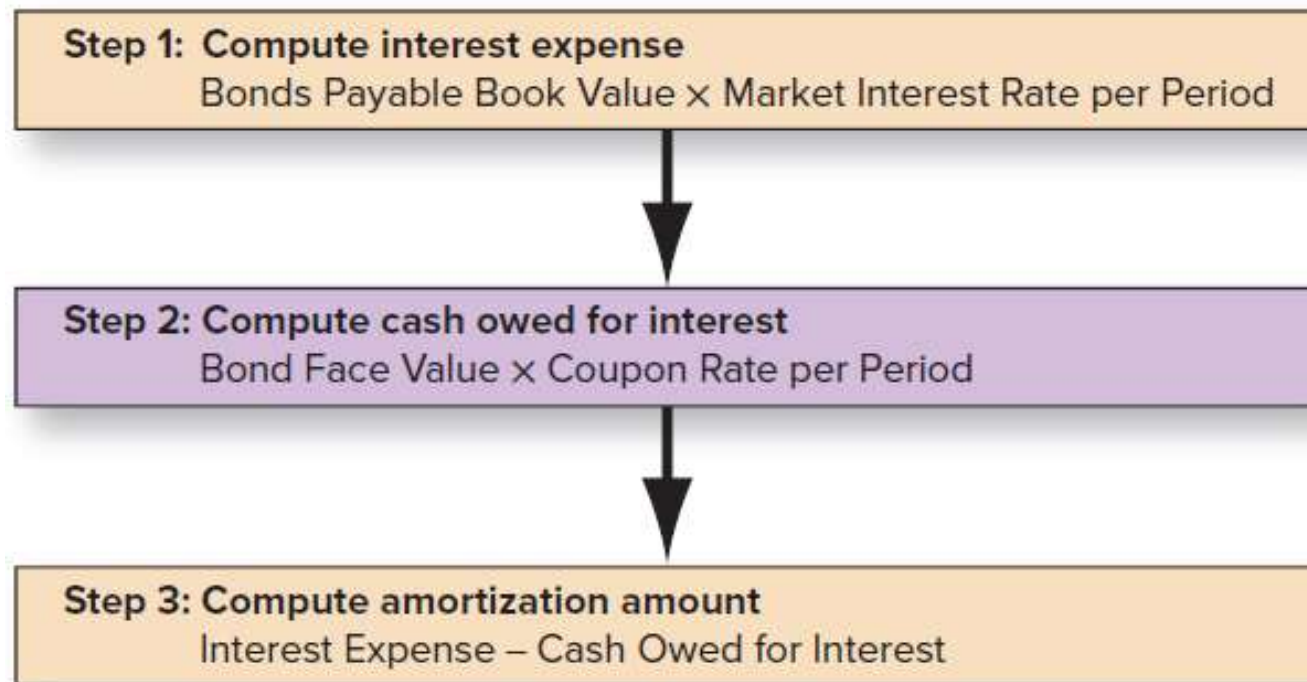
While Amazon received only \$96,535 when it sold the bonds, it must repay \$100,000 when the bonds mature.

Bonds Issued at a Discount (4 of 4)

- While Amazon received only \$96,535 when it sold the bonds at a discount, it must repay \$100,000 when the bonds mature.
 - The \$3,465 of additional cash that Amazon must pay at maturity is incrementally reflected in interest expense and added to the book value of the liability each period over the life of the bond.
- This causes interest expense to reflect the **market interest rate** (also referred to as the **effective interest rate**), while the cash being paid to bondholders each period reflects the **coupon rate**.
- The accounting method used to record interest expense is referred to as the **effective-interest amortization** method.

Reporting Interest Expense on Bonds Issued at a Discount Using Effective-Interest Amortization (1 of 3)

- Under the effective-interest amortization method, a company computes interest expense by multiplying the bonds payable book value times the market rate of interest on the date of issuance.
- When bonds are issued at a discount interest expense each period is greater than the cash owed for interest each period.
- The difference between interest expense and the cash owed for interest is the amount of the bond discount amortized during the period.



Reporting Interest Expense on Bonds Issued at a Discount Using Effective-Interest Amortization (2 of 3)

- The cash owed for interest is computed by multiplying the bond's face value (\$100,000) by the coupon rate per period ($10\% \times \frac{1}{2}$ year).
- Thus, Amazon owes bondholders cash of \$5,000 each June 30 and again on December 31. The first interest payment on the bonds is made on June 30, 2025.

Step 1: Interest expense: $\$96,535 \times (0.12 \times \frac{1}{2} \text{ year}) = \$5,792$

Step 2: Cash owed for interest: $\$100,000 \times (0.10 \times \frac{1}{2} \text{ year}) = \$5,000$

Step 3: Amortized amount: $\$5,792 - \$5,000 = \$792$

The journal entry Amazon would enter is:

Effective-interest amortization causes these amounts to change each period because the book value of the bonds changes each period.

June 30, 2025	Debit	Credit
Interest expense (+E, -SE)	5,792	
Bond discount (+L)		792
Cash (-A)		5,000

Assets	=	Liabilities	+	Stockholders' Equity
Cash -5,000		Bond discount +792		Interest expense (+E) -5,792

Bonds payable book value: **\$97,327** (\$100,000 beginning balance - \$3,465 + \$792)

Reporting Interest Expense on Bonds Issued at a Discount Using Effective-Interest Amortization (3 of 3)

- Interest expense for the next interest period must reflect the change in the bonds payable book value.
- Interest expense for the second half of 2025 is calculated as follows:
 - Multiply the bonds payable book value on June 30, 2025 (\$97,327) by the market rate of interest per period:
 $\$97,327 \times (12\% \times \frac{1}{2} \text{ year}) = \$5,840$
 - With interest expense equal to \$5,840 and cash owed for interest equal to \$5,000, the amount of the bond discount amortized on December 31, 2025, is \$840.

December 31, 2025	Debit	Credit
Interest expense (+E, – SE)	5,840	
Bond discount (+L)		840
Cash (– A)		5,000

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	–5,000		Bond discount	+840		Interest expense (+E)	–5,840
Bonds payable book value: \$98,167 (\$100,000 beginning balance – \$3,465 + \$792 + \$840)							

Bond Discount Amortization Schedule

Interest expense for December 31, 2025, is more than interest expense for June 30, 2025. Amazon borrowed more money during the second half of the year because of the unpaid interest accrued during the first half of the year.

File Home Insert Page Layout Formulas Data Review View Automate				
Market rate/period = 12% / 2 periods				Journal Entry
	B	C	D	E
1	BOND AMORTIZATION SCHEDULE: ISSUED AT A DISCOUNT			
2	(a)	(b)	(c)	(d)
3	Face × Coupon Rate/Period	Beginning of Period Book Value × Market Rate/Period	Absolute Difference between (a) and (b)	Beginning of Period Book Value + (c)
4	Credit	Debit	Credit	
5	Cash	Interest Expense	Bond Discount	Bonds Payable
6	Owed for Interest	Incurred	Amortized	Book Value
7	Date			
8	1/1/2025			96,535
9	6/30/2025	5,000	5,792	792
10	12/31/2025	5,000	5,840	840
11	6/30/2026	5,000	5,890	890
12	12/31/2026	5,000	5,943	943
13				100,000
14				
Book value moves toward face value				

Learning Objective 10-5

10-5 Report bonds payable and interest expense for bond securities issued at a premium.

Bonds Issued at a Premium (1 of 3)

When the **market interest rate** is less than the bond's coupon rate, the bonds sell at a **premium**.

Bonds Issued at a Premium



Bonds Issued at a Premium (2 of 3)

On January 1, 2025, Amazon issues bonds with a coupon rate of 10 percent and a face value of \$100,000. The **market interest rate is now 8 percent**. The bonds start accruing interest on January 1, 2025, and will pay interest each June 30 and December 31. The bonds mature in two years on December 31, 2026. The coupon rate (10 percent) is less than the market interest rate (12 percent) on the date of issuance, so the bonds sell at a **premium**.

	Present Value
Single principal payment at maturity: $\$100,000 \times 0.85480$	\$ 85,480
+ Annuity cash interest payment: $\$5,000 \times 3.62990$	18,150
Issue (sale) price of bonds	<u>\$103,630</u>

INPUTS USING:

Table E.1	Table E.2	Calculator	Excel
Interest rate (i) = 4%	Interest rate (i) = 4%	Rate (i) = 4	Rate (i) = 0.06
Periods (n) = 4	Periods (n) = 4	Periods (n) = 4	Nper (n) = 4
Factor = 0.85480	Factor = 3.62990	Pmt = $-\$5,000$	Pmt = $-\$5,000$
		FV = $-\$100,000$	Fv = $-\$100,000$

Bonds Issued at a Premium (3 of 3)

Accounting for bonds issued at a premium is similar to accounting for bonds issued at a discount.

Companies can explicitly use a bond premium account in their journal entries or implicitly keep track of the premium amount.



The bond payable book value is equal to the bond's face value plus the unamortized bond premium.

The journal entries to record the sale of Amazon's bonds issued at a premium are:

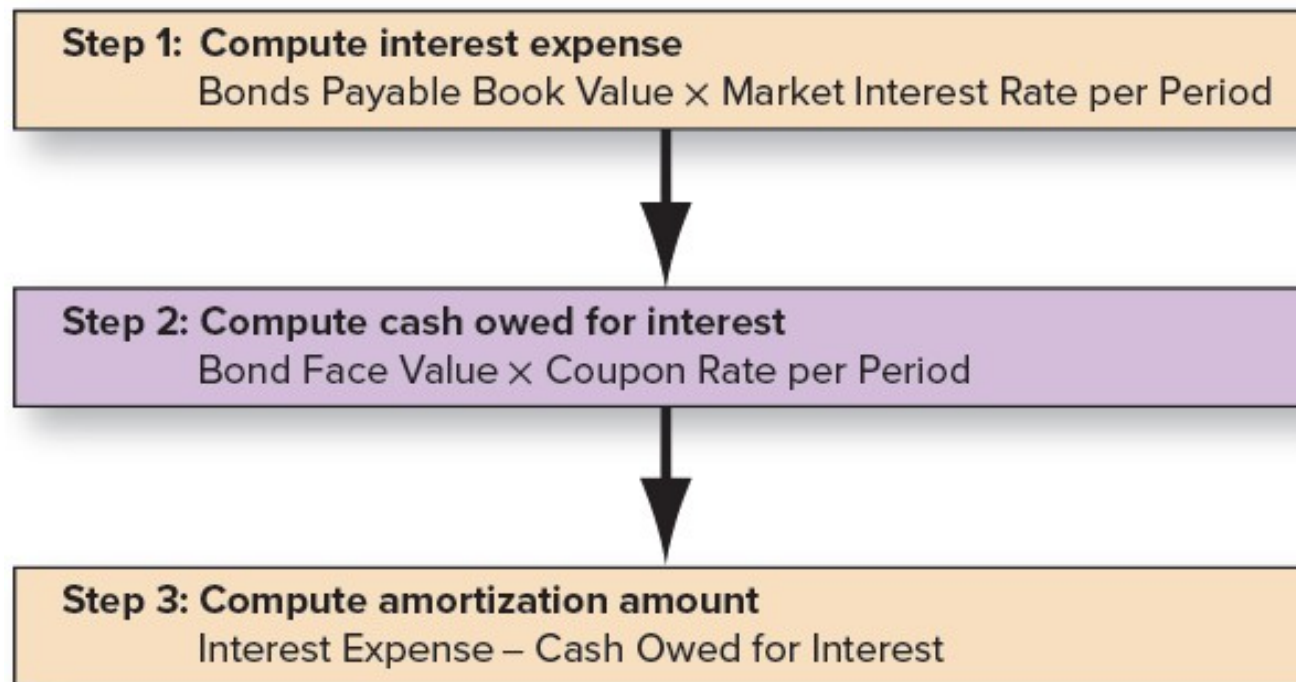
January 1, 2025	Debit	Credit
Cash (+A)	103,630	
Bond premium (+L)		3,630
Bonds payable (+L)		100,000

Assets		=	Liabilities		+	Stockholders' Equity
Cash	+103,630		Bonds payable	+100,000		
			Bond premium	+3,630		

Bonds payable book value: **\$103,630** (\$100,000 beginning balance + \$3,630)

Reporting Interest Expense on Bonds Issued at a Premium Using Effective-Interest Amortization (1 of 2)

Under the effective-interest amortization method, a company computes interest expense in a given period by **multiplying the bonds payable book value times the market rate of interest** on the date of issuance.



Reporting Interest Expense on Bonds Issued at a Premium Using Effective-Interest Amortization (2 of 2)

Step 1: Interest expense: $\$103,630 \times (0.08 \times \frac{1}{2} \text{ year}) = \$4,145$

Step 2: Cash owed for interest: $\$100,000 \times (0.10 \times \frac{1}{2} \text{ year}) = \$5,000$

Step 3: Amortized amount: $\$4,145 - \$5,000 = -\$855$

The journal entry Amazon would enter is:

June 30, 2025	Debit	Credit
Interest expense (+E, -SE)	4,145	
Bond premium (-L)	855	
Cash (-A)		5,000

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	-5,000		Bond premium	-855		Interest expense (+E)	-4,145
Bonds payable book value: \$102,775 (\$100,000 beginning balance + \$3,630 - \$855)							

The amortization of a discount **increases** the book value of the liability and the amortization of a premium **decreases** it. Both serve the same purpose: to bring the liability to the bond's face value at the maturity date.

Bond Premium Amortization Schedule

The following schedule illustrates the amortization of a premium over the life of a bond.

File Home Insert Page Layout Formulas Data Review View Automate				
Market rate/period = 8% / 2 periods		X ✓ f_x v	Journal Entry	
	B	C	D	E
1	BOND AMORTIZATION SCHEDULE: ISSUED AT A PREMIUM			
2	(a)	(b)	(c)	(d)
3	Face ×	Beginning of Period	Absolute	Beginning of Period
4	Coupon Rate/Period	Book Value ×	Difference between	Book Value – (c)
5		Market Rate/Period	(a) and (b)	
6	Credit	Debit	Debit	
7	Cash	Interest Expense	Bond Premium	Bonds Payable
8	Owed for Interest	Incurred	Amortized	Book Value
9	1/1/2025			103,630
10	6/30/2025	5,000	4,145	102,775
11	12/31/2025	5,000	4,111	101,886
12	6/30/2026	5,000	4,075	100,962
13	12/31/2026	5,000	4,038	100,000
14				

Book value moves toward face value

Journal Entry to Retire the Bonds at Maturity

Regardless of whether a company issues bonds at par, at a discount, or at a premium, the company will enter the same journal entry when it retires the bonds at maturity.

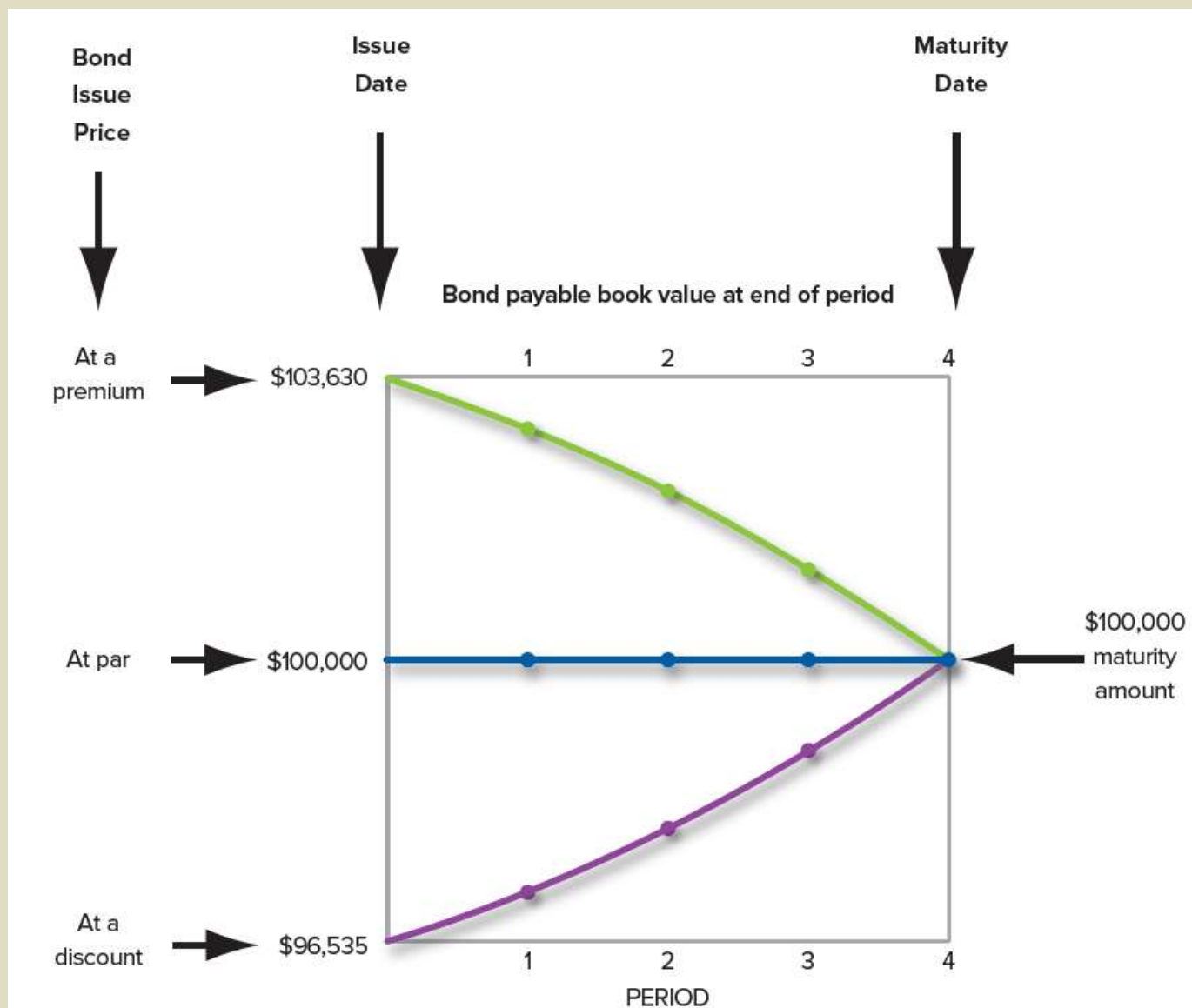
For our Amazon example, the journal entry at 12/31/2026 would be:

December 31, 2026	Debit	Credit
Bonds payable (–L)	100,000	
Cash (–A)		100,000

Assets		=	Liabilities		+	Stockholders' Equity
Cash	–100,000		Bonds payable	–100,000		

Exhibit 10.3

The Change in the Book Value of a Bond Over Time



Zero Coupon Bonds



FINANCIAL ANALYSIS

Zero coupon bonds do not pay periodic cash interest. Investors will price the bond so that they earn the market rate of interest.

Because there is no periodic cash paid for interest:

$$\text{PV of the Principal} = \text{Issue Price of the Bonds}$$

- A bond with a zero coupon interest rate is simply a deeply discounted bond that will sell for substantially less than the amount due at maturity (its face value).
- Accounting for a zero coupon bond is no different than accounting for other bonds issued at a discount.

However, the amount of **the discount is much larger.**

Learning Objective 10-6

10-6 Report the early retirement of bond securities.

Early Retirement of Bonds

Some bonds have a **call** feature that allows the issuing company to call (retire) the bonds early. A call feature most often requires the issuing company to pay investors an amount greater than the bond's face value to retire the bonds before their maturity date. The amount paid is often stated as a percentage of the bond's face value.

Cash Paid to Retire Bonds < Book Value = **GAIN**

Cash Paid to Retire Bonds > Book Value = **LOSS**

In some cases, a company may elect to retire bonds early by purchasing them on the **open market**, just as an investor would. This approach is necessary when the bonds do not have a call feature. Even when the bonds have a call feature, retiring them by purchasing them on the open market is attractive when the cost of doing so is less than the cost of paying the call premium.

Journal entry to retire bonds

Assume the following:

- Book value of Amazon's bonds was \$101,000 at the time the bonds were retired.
- Amazon paid \$102,000 to retire the bonds.
- Amazon would recognize a loss on bond call of \$1,000 on its income statement.

The journal entry would be:

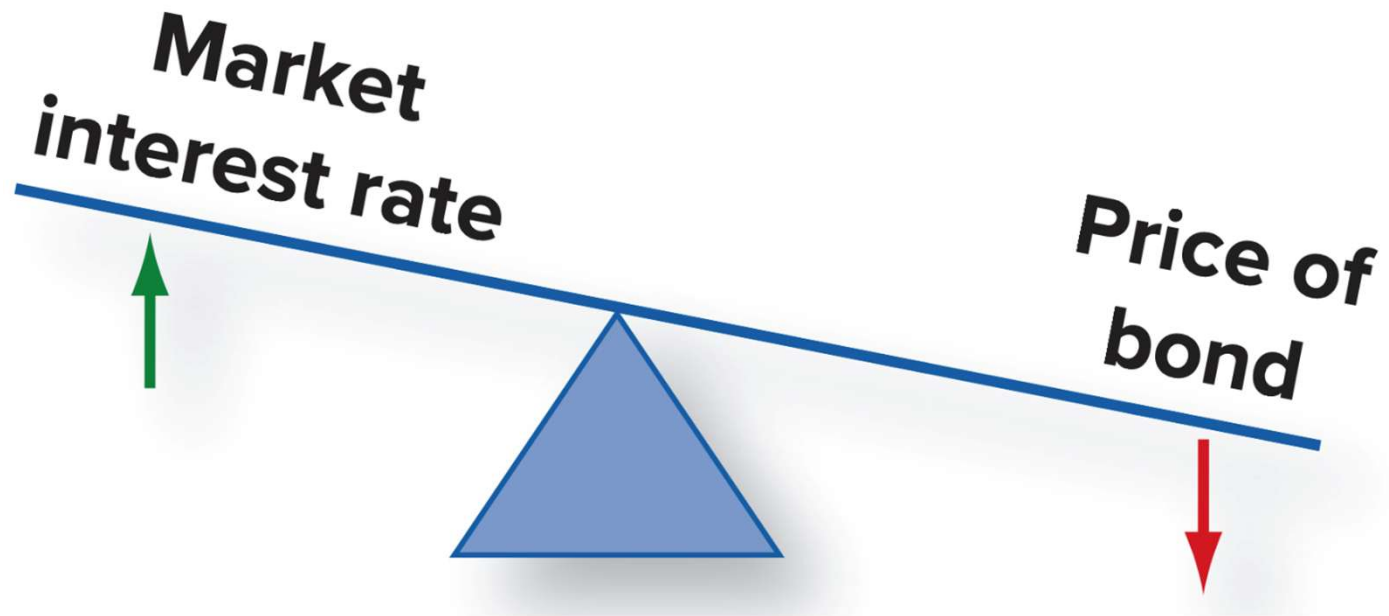
Date bonds are retired	Debit	Credit
Bonds payable (–L)	100,000	
Bond premium (–L)	1,000	
Loss on bond call (+E, –SE)	1,000	
Cash (–A)		102,000

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	–102,000		Bonds payable	–100,000		Loss on bond call (+E)	–1,000
			Bond premium	–1,000			

Price of Bonds versus Market Interest Rates

Bond prices move in the opposite direction of interest rates.

If interest rates go up, bond prices fall, and vice versa. If interest rates go up enough, a company may decide that it makes good economic sense to retire its bonds early by purchasing them on the open market.



Reporting Interest Expense Using Straight-Line Amortization

- **GAAP** requires that companies use the **effective-interest method** to **amortize bond discounts and bond premiums**. GAAP permits companies to use straight-line amortization when results do not materially differ from results computed using the effective-interest method.
- With **straight-line amortization**, a company takes the total amount of the discount or the premium at issuance, divides it by the number of periods in the bond's life, and amortizes that amount each period.

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114

4 periods =
2 years × 2 times
per year

Journal Entry

Coupon rate/period =
10% / 2 periods

BOND AMORTIZATION SCHEDULE: ISSUED AT A PREMIUM (STRAIGHT-LINE AMORTIZATION)

	(a)	(b)	(c)	(d)
	Face × Coupon Rate/Period	(a) – (c)	Bond Premium/ Number of Periods	Beginning of Period Book Value – (c)
	Credit	Debit	Debit	
	Cash	Interest Expense	Bond Premium	Bonds Payable
	Owed for Interest	Incurred	Amortized	Book Value
1/1/2025				103,630.00
6/30/2025	5,000	4,092.50	907.50	102,722.50
12/31/2025	5,000	4,092.50	907.50	101,815.00
6/30/2026	5,000	4,092.50	907.50	100,907.50
12/31/2026	5,000	4,092.50	907.50	100,000.00

Book value moves
toward face value

Learning Objective 10-7

10-7 Compute and analyze the debt-to-equity ratio



This ratio shows the relationship between the amount of capital provided by owners and the amount provided by creditors.

In general, a high ratio indicates that a company relies heavily on debt financing relative to equity financing. This increases the risk that a company may not be able to meet its contractual financial obligations during a business downturn.

$$\text{Debt-to-Equity} = \frac{\text{Total Liabilities}}{\text{Total Stockholders' Equity}}$$

The 2023 ratio for Amazon is

$$\frac{\$325,975}{\$201,875} = 1.61$$

COMPARISON OVER TIME		
Amazon		
	2022	2023
	2.17	1.61

COMPARISON WITH COMPETITORS	
eBay	Walmart
2023	2023
2.38	1.9

Learning Objective 10-8

10-8 Explain how bond securities are reported on the statement of cash flows

Bond Transactions and the Statement of Cash Flows (10-52)

FOCUS ON CASH FLOWS



Transaction	Where Reported in Cash Flow Statement
Cash paid for interest	– Operating Activities
Cash received when bonds are issued	+ Financing Activities
Cash paid when bonds mature or are redeemed	– Financing Activities

Payment of interest under GAAP is an operating cash outflow.

Interest expense is reported on the income statement and is a component of net income. As a result, GAAP requires interest payments be reported as operating activities on the statement of cash flows.

Analysts are particularly interested in the financing activities section of the statement of cash flows because it provides important insights about the capital structure of a company.

Bond Transactions and the Statement of Cash Flows (2022)

FOCUS ON CASH FLOWS



Excerpt from **Amazon's** cash flow statement showing its financing activities:

AMAZON.COM, INC.			
CONSOLIDATED STATEMENTS OF CASH FLOWS (Partial)			
in millions			
	Year Ended December 31,		
	2021	2022	2023
...			
FINANCING ACTIVITIES:			
Common stock repurchased	—	(6,000)	—
Proceeds from short-term debt	56	41,553	18,129
Repayments of short-term debt and other	(1,554)	(37,554)	(25,677)
Proceeds from long-term debt	1,554	21,166	—
Repayments of long-term debt	(1,554)	(1,258)	(3,676)
Principal repayments of finance leases	(11,163)	(7,941)	(4,384)
Principal repayments of financing obligations	(162)	(248)	(271)
Net cash provided by (used in) financing activities	6,291	18,867	(15,879)

Chapter Supplement: Accounting for Bonds without a Discount Account or Premium Account

For financial reporting purposes, it is not necessary to use a discount (or premium) account when recording the sale of bonds. The Chapter Supplement duplicates the journal entries in this chapter without using the bond discount or bond premium accounts.

For example, the journal entries to record the sale of bonds issued at a discount without using a discount account is:

January 1, 2025	Debit	Credit
Cash (+A)	96,535	
Bonds payable (+L)		96,535

The journal entry to record interest expense, without the discount account:

June 30, 2025	Debit	Credit
Interest expense (+E, –SE)	5,792	
Bonds payable (+L)		792
Cash (–A)		5,000